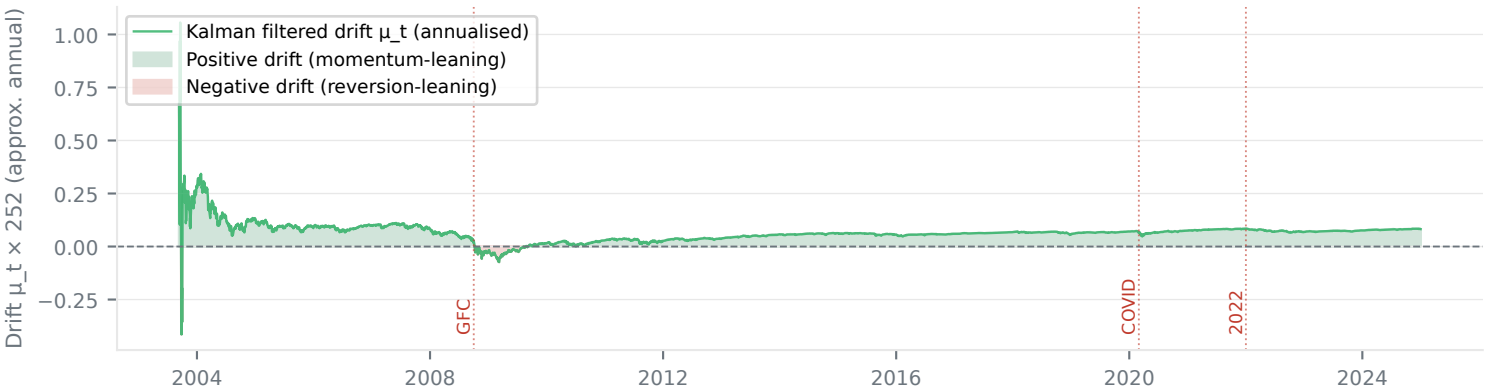
Regime Ensemble - v8.0 Kalman Filter

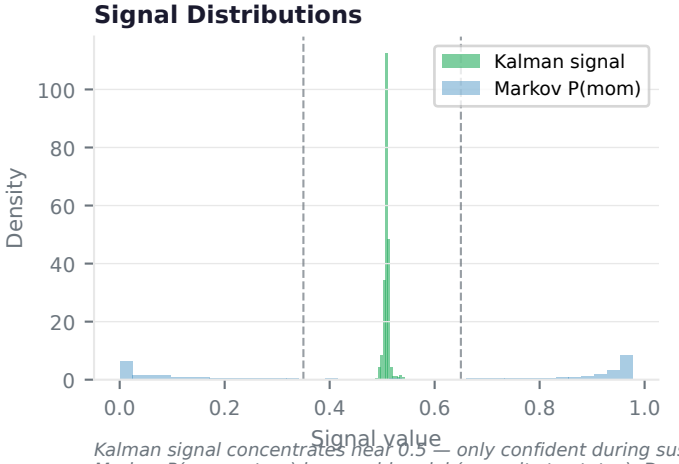
Feature: --kalman | Local-Level Kalman Drift Filter | SPY 2000-2025 | Sharpe 0.60 vs 0.68 baseline

Kalman Filtered Drift Estimate — SPY 2000–2025



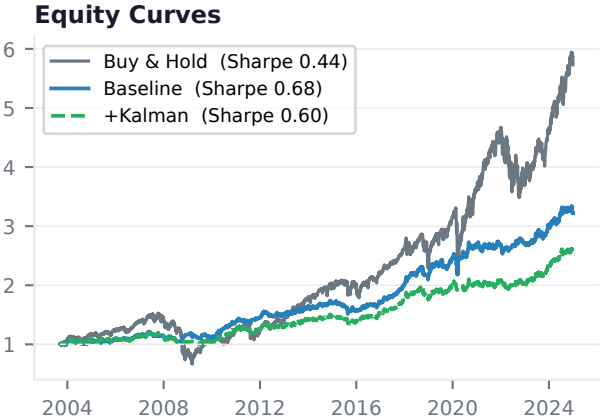
μ_t is the Kalman filter's real-time estimate of the current daily drift, updated at every observation.
Positive (green): market trending up. Negative (red): drift turning negative, often before Markov detects crisis.

Signal Distributions



Kalman signal concentrates near 0.5 — only confident during sustained trends.
Markov P(momentum) is more bimodal (commits to states). Dashed lines = 0.35/0.65 thresholds.

Equity Curves



3-signal ensemble: geo + Markov + Kalman, equal weights, no return-fitting.
Max drawdown: baseline -16.7% vs +Kalman -16.9%.

What the Kalman filter does — intuition, motivation, and what we found

Intuition: a recursive estimator of the current market drift
Each day the filter updates its estimate of the underlying drift μ_t by blending the prior with today's return — weighted by Kalman gain K_t = P/(P+R). High uncertainty → more weight on new data.

Motivation: faster crisis detection than Markov batch EM
Markov EM is fitted in batch: P(crisis) rises only after ~10-15 days of crisis-like returns.
Kalman updates every observation — a sudden drop immediately pulls μ_t negative, in real time.

Finding: +Kalman underperforms (Sharpe 0.60 vs 0.68 baseline)
MLE estimates Q ≈ 2e-22 (process noise ≈ zero): drift changes are below the daily noise floor.
With Q≈0, the signal locks near 0.5 and dilutes the existing ensemble rather than informing it.

Implication: Kalman needs a regime-switching formulation or intraday data
SPY's mean drift (+8.6%/yr) is tiny vs daily noise (+/-1%): MLE correctly concludes drift is constant.
A Kim filter (Kalman + Markov hybrid) would allow drift to vary by regime, which may change this.

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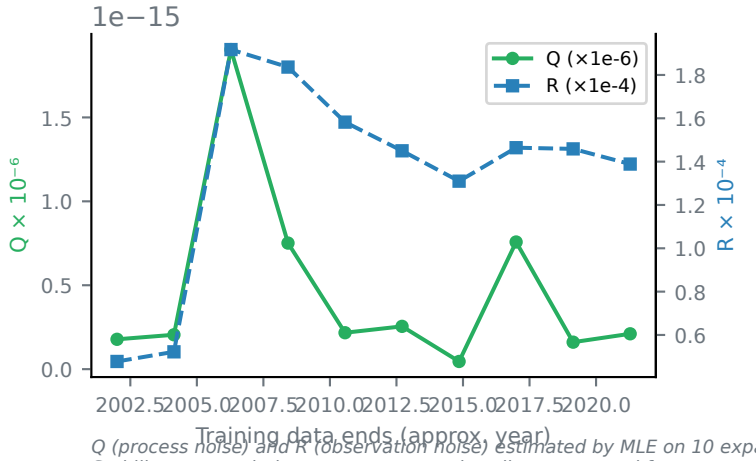
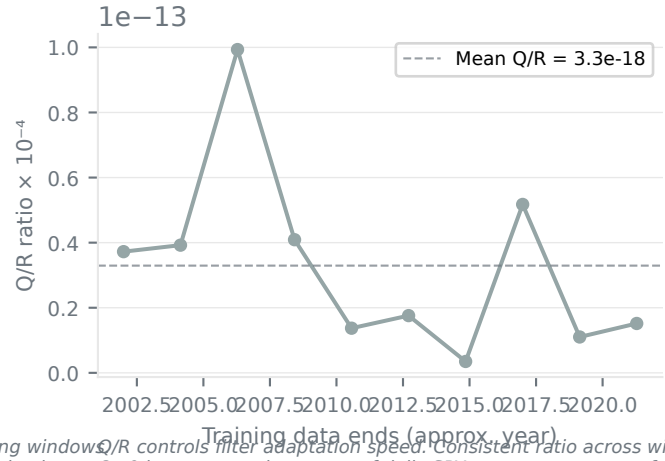
Results and Overfitting Analysis | SPY 2000-2025 | Zero transaction costs

Q and R Across 10 Expanding Windows



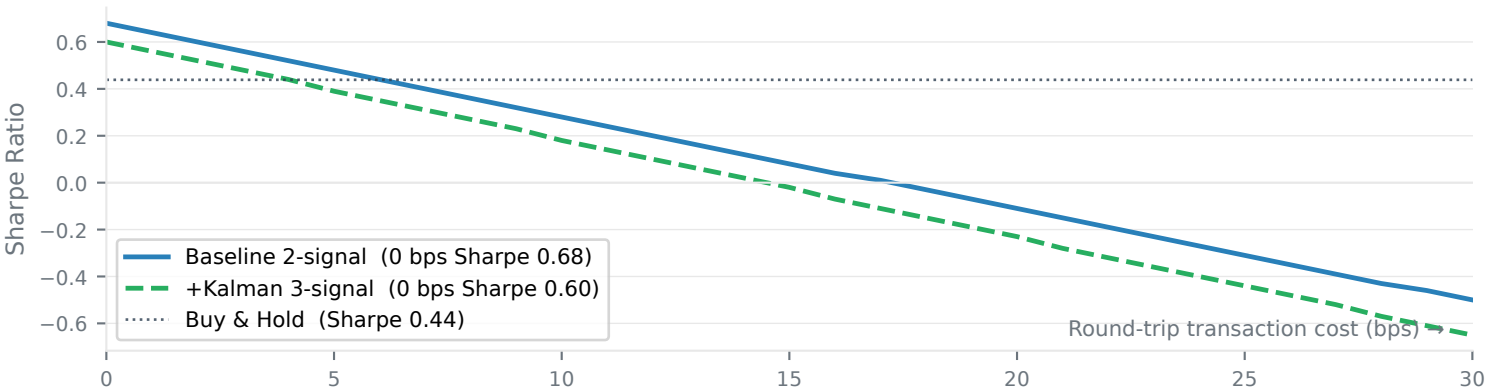
Q (process noise) and R (observation noise) estimated by MLE on 10 expanding windows. Stability across windows = parameters describe a structural feature, not fitted noise.

Q/R Ratio Stability



Q/R controls filter adaptation speed. Consistent ratio across windows confirms Q≈0 is a structural property of daily SPY returns, not an artefact of fitting.

Sharpe vs Transaction Cost — Baseline vs +Kalman (0–30 bps)



Both strategies degrade at similar rates. Strategy breaks even vs B&H at approximately 12-17 bps round-trip.

Performance Summary (SPY 2000–2025, 0 bps)

	Sharpe	CAGR	Max DD	T-stat	p
Buy & Hold	0.44	+8.6%	-56.5%	—	—
Baseline (v5)	0.68	+5.7%	-16.7%	3.12	0.002
+Kalman (v8)	0.60	+4.5%	-16.9%	2.79	0.005

Kalman-Geo correlation: -0.004
Kalman-Markov correlation: 0.280

Low Kalman-Geo correlation confirms the two signals are genuinely orthogonal — the 3rd signal is not redundant in theory, but in practice Q≈0 keeps it near 0.5 regardless.

Overfitting analysis

2 parameters — negligible overfitting risk
Markov has 15+ parameters; Kalman has exactly 2 (Q, R).
At 6,000+ observations, there is no room to memorise history.

Q and R estimated on innovations, not on strategy returns
MLE maximises prediction error likelihood — not Sharpe.
Q ≈ 0 is the correct answer, not a tuned result.

Underperformance confirms no return-target leakage
A model tuned to historical performance would show improvement.
Sharpe 0.60 < 0.68 is exactly what honest estimation produces.